

第2课 局域异步通信



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一、单项选择题

1. In RS-232-C, () voltage corresponds to logical 1.
A. increasing B. decreasing C. negative D. positive
2. RS-232 DB9 uses () transmission during the communication.
A. simplex B. half duplex C. full duplex D. none
3. The RS-232 standard sends each bit of the character, and follows each character with an idle period () bit(s) long.
A. no B. one C. at least one D. at most one.
4. Which is correct about RS-232-C?
A. The connection must be less than 15 feet long.
B. The voltage ranges from -50 volts to +50 volts.
C. A sender transmit an extra 1 bit (called a start bit) before transmitting a character
D. A sender must can send characters without leaving the line idle between them.
5. Which of the following about RS-232-C is correct?
A. RS-232-C specifies that a sender transmits a start bit before transmitting the bits of a character, and a stop bit is appended to each character.
B. When it finishes transmission, the sender leaves the wire with a positive voltage until another character is ready for transmission.
C. RS-232-C uses voltage ranging from -5 volts to +5 volts.
D. Negative voltage corresponds to logical 0, while positive voltage corresponds to logical 1.
6. Which of the following about RS-232-C is WRONG?
A. Each data item represents one character in keyboard.
B. Once transmission begin, a sender transmits a start bit, all bits of the characters, and there is one stop bit at the end of transmission.

- C. The connection must be less than 50-feet long.
- D. The hardware can be configured to control the exact number of bits per second.
7. Which of the following about RS-232-C is WRONG?
- A. It is used for asynchronous, serial communication over short distances.
- B. Negative voltage corresponds to logical 0, while positive voltage means logical 1.
- C. A sender transmits all bits of the byte one after another with no delay between them.
- D. RS-232-C uses voltage ranging from -15 volts to +15 volts.
8. Nyquist Theorem tells a theoretical bound on the maximum rate is related to ().
- A. baud rate B. coding error C. signal intensity D. noise intensity

二、简答题

9. 有哪几种串行传输的类型？为下面几种场景选择正确的类型。（1）光纤干线。（2）计算机键盘连接。（3）视频传输，如：项目一（或实验一）利用可见光近距离传输。（4）RS232C。
10. 按信号传输方向划分，有哪几种？为下面几种场景选择正确的类型。（1）语音对讲机。（2）项目一（或实验一）。（3）RS232C DB9。
11. 项目一（或实验一），如果不经手机翻拍，直接用源视频解码。假设屏幕最大的分辨率是 1920×1080 像素，帧率是 30fps，颜色中 RGB 分量分别为 0~255。不考虑视频压缩产生的失真、定位点占用的空间等损失。此时，如何按奈奎斯特定理，计算其最大数据速率？
12. 假定用 38,400 baud 波特率的 RS-232 方式发送了 10000 个 7 位字符，则传输需要多少时间？（每个字符都有一个起始位和一个终止位）
13. 以 RS232 标准为例说明物理层接口的主要特征。
14. 请以 RS232-C 传输 8 位字符“X”（其 ASCII 码为 88）为例，画出电压随时间变化示意图。标明横轴和纵轴的刻度，并注明每个时刻传输的内容。
15. 绝大多数的 RS-232 硬件允许所在的计算机设定数据传输速率及停止位的个数。若发送方使用两位停止位，而接收方使用一位停止位，数据传输是否正确？如果正确，使用额外的停止位的不利之处是什么？

三、程序设计题

16. 请编写一个程序，以字节大端序、位小端序，输入一个字母，模拟 RS232C，以 7-bit 编码方式，包括空闲位、起始位和终止位，输出电压值序列。

接口：`int rs232c_encode(double *volts, int volts_size, const char *msg, int size); int rs232c_decode(char *msg, int size, const double *volts, int volts_size);`。